Beekeeping and Pollination Studies

Science

May, 2026

Beekeeping and Pollination Studies (A14405)

Short Title:	Pollinators		
Faculty	Science and Computing		
Department	Science		
Cluster	Horticulture		
Author	DFLANAGAN		
Credits	5	**Level:**	Intermediate

Description of Module / Aims

This module aims to equip students with the knowledge and skills associated with apiculture, commercial pollination of crops and natural pollination services.

Programmes

HORT-0046	BSc in Horticulture (WD_SHORT_D)	3 / 6 / E
HORT-0046	BSc in Horticulture (WD_SHORT_DX)	3 / 6 / E

Indicative Content

- The life cycle of the honey bee, bumble bee, other pollinators and their colonies
- Apiary site selection criteria, annual management programmes.
- Explanation of beekeeping equipment, assembly, and inspection of hives
- Nutrition and food sources for pollinators
- Criteria and methods of breeding
- Pest, disease, poisons and habitats of bees and other pollinators
- Key legislation, directives and action plans to protect pollinating species

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Associate pollinator biology, pollinator health and potential challenges.
2. Produce a plan outlining the management of a commercial apiary and protected pollinators.
3. Demonstrate routine skills associated with managing honey bees.
4. Compare the variety of forage crops for pollinators.
5. Compare attributes when selecting and breeding pollinators.
6. Relate the Sustainable Use of Pesticide Directive, and the current National Pollinator Action Plan to improved outcomes for pollinators.

Learning and Teaching Methods

- Lectures
- Demonstration
- Field practical

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,4,5,6
Continuous Assessment	50%	
Assignment	20%	2
Practical	30%	3

Assessment Criteria

<40%: Not able to identify 40% of pollinating insect castes and equipment, unable to describe management practices and applications, unable to assemble hive, and unable to describe suitable apiary sites.

40%-49%: Can identify at least 40% of pollinating insect castes and equipment, can describe management practices and application in industry, able to assemble hive, and describe suitable apiary sites.

50%-59%: All of above and can correctly identify at least 60% of pollinating insect castes and equipment, can describe suitable apiary sites and forage requirements.

60%-69%: All of the above with at least 70% of pollinating insect castes and equipment identified correctly.

70%-100%: All of the above and at least 80% of pollinating insect castes and equipment identified correctly.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Essential Material

- Hooper, T. _Guide to Bees and Honey_. UK: Northern Bee Books, 2010.

Supplementary Material

- "National Bee Unit." http://www.nationalbeeunit.com/
- "National Biodiversity Data Centre." http://www.biodiversityireland.ie/
- McMullan, J. _Having Healthy Honeybees_. Ireland: The Federation of Irish Beekeepers' Associations, 2012.
- Wilmer, P. _Pollination and Floral Ecology_. America: Princeton University Press, 2011.

Requested Resources

- Lecture Room: Loose Seated